

Power-Residue Tiles as Finite Arithmetic Objects

Visual Taxonomy, Coordinate Laws, and Base-Covariant Structure

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Abstract

For each integer modulus $d \geq 1$, the *power-residue tile* is the finite two-coordinate object

$$T_d(x, k) = x^k \pmod{d},$$

with row coordinate $x \in \mathbb{Z}/d\mathbb{Z}$ and exponent coordinate $k \geq 1$. In base b , the familiar digital-root power table is the display convention associated to the modulus $d = b - 1$. This paper studies the tile itself as a finite arithmetic object, rather than as a pattern table or visualization aid.

The central claim is restrained and algebraic: the visible structure of a digital-root power tile is completely governed by finite residue arithmetic. Unit rows are periodic, with periods dividing the Carmichael exponent $\lambda(d)$. Inverse rows are rotations of one another inside their unit cycles. Non-unit rows are governed by prime-power valuation growth and Chinese-remainder decomposition; they either reach the zero class in nilpotent components or become eventually periodic in the surviving CRT components. Columns act as exponent maps, while rows exhibit orbit flow.

The paper introduces a coordinate taxonomy for the tile: row coordinate, exponent coordinate, displayed residue, gcd-depth, prime-power valuation profile, quotient-unit orbit, CRT position, row family, column action, and tile-period codebook. The base-10 9×9 DRPT tile is presented as the canonical visual witness, not as decoration: it is the full finite object for $d = 9$ under the conventional digital-root display. A 900-base audit is included as a reproducibility certificate, not as a substitute for proof. The results are elementary, finite, and formalizable in ZFC.

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1 Introduction

A digital-root power table is often encountered as a small recreational table: rows are bases, columns are powers, and entries are reduced to digital roots. In base 10, the familiar table displays the digital roots of n^k .

This paper proposes a different reading. The table is not merely a pattern. It is the finite visual trace of exponentiation in a residue ring.

For base b , the induced modulus is

$$d = b - 1.$$

The base- b digital-root behavior of an integer is determined by its residue modulo d . Thus a base- b digital-root power table is a display of the map

$$(x, k) \mapsto x^k \pmod{b-1}.$$

This object will be called a *power-residue tile*. When displayed through the base- b digital-root convention, it will be called a *Digital Root Power Table*, or DRPT.

The motivating observation is simple: the one-dimensional number line orders residues, but the tile reveals how residues act under exponentiation. The line shows a sequence. The tile shows all row orbits and all exponent actions at once.

A displayed value is therefore not the whole datum. A cell has position:

$$(x, k, T_d(x, k)).$$

The value 3 in one cell is not structurally identical to another displayed 3 unless row coordinate, exponent coordinate, gcd-depth, quotient-unit position, and CRT context agree. In the tile, location matters.

Related work

Iteration of monomial maps $x \mapsto x^k$ on residue rings has a long history, but the literature has overwhelmingly framed the question dynamically: fix an exponent k , regard $x \mapsto x^k$ as a self-map of R_d , and study the resulting functional graph (cycle structure, tail length, component count). Rogers [9] initiated this perspective for the squaring map on prime fields. Vasiga and Shallit [11] sharpened the analysis for the squaring map over $\text{GF}(p)$, including precise enumeration of periodic points. Chou and Shparlinski [2] extended the framework to repeated exponentiation $x \mapsto x^e$ with arbitrary fixed exponent e , removing dependence on the Extended Riemann Hypothesis. More recent work studies the iteration graph $\mathcal{U}(p^k)$ of $x \mapsto x^2$ on prime-power moduli [10]. Properties of digital-root power sequences in base 10 are recorded expositively by Izmirli [8].

The object studied here is the static dual of those dynamical objects. Rather than fix k and iterate, this paper fixes the algebraic exponentiation map $(x, k) \mapsto x^k \bmod d$ and tabulates it on the full coordinate plane $R_d \times \mathbb{N}_{\geq 1}$. The unit cycles, inverse rotations, and absorbing strata that appear as components of a Vasiga–Shallit functional graph appear here as rows of a single static incidence object. The two perspectives are equivalent in content but display different information: a functional graph emphasizes orbit topology under one map; the tile emphasizes simultaneous row periods, inverse-pair symmetry, and non-unit valuation flow across all exponents at once. The base-covariant construction (Section 14) and the unified codebook bound (Theorem 12) are stated in this static frame.

Background material on Carmichael’s function $\lambda(d)$, the Chinese Remainder Theorem, and prime-power valuations is standard; see [1, 7]. Closely related Fourier-analytic and spectral-operator material on $\mathbb{Z}/N\mathbb{Z}$, including explicit kernel norms and admissibility of averaging operators, appears in [3, 4].

2 The object

2.1 Base, modulus, and residue ring

Fix an integer base $b \geq 2$. Define

$$d = b - 1.$$

The relevant finite ring is

$$R_d = \mathbb{Z}/d\mathbb{Z}.$$

All structural statements in this paper are made in R_d . The digital-root display is a presentation layer.

Definition 1 (Power-residue tile). For $d \geq 1$, define the power-residue tile

$$T_d : R_d \times \mathbb{N}_{\geq 1} \rightarrow R_d$$

by

$$T_d(x, k) = x^k \pmod{d}.$$

For a finite exponent horizon K , define the finite tile

$$T_{d,K} : R_d \times \{1, \dots, K\} \rightarrow R_d.$$

This is a finite object with dK cells.

2.2 Digital-root display convention

For base b , the standard digital-root display uses the alphabet $\{1, 2, \dots, b-1\}$, with the class $0 \pmod{b-1}$ often displayed as $b-1$. In base 10, this means the residue class $0 \pmod{9}$ is displayed as 9.

Define the display map

$$\delta_d(r) = \begin{cases} d, & r \equiv 0 \pmod{d}, \\ r, & 1 \leq r \leq d-1. \end{cases}$$

The displayed DRPT tile is

$$D_d(x, k) = \delta_d(x^k \pmod{d}).$$

The mathematics occurs in R_d . The display is a human-readable rendering.

3 The canonical base-10 tile

The base-10 digital-root power tile has $d = 9$. Using the conventional display $0 \mapsto 9$, the 9×9 tile is shown in Figure 1.

		exponent k									
		1	2	3	4	5	6	7	8	9	
row n	1	1	1	1	1	1	1	1	1	1	identity row
	2	2	4	8	7	5	1	2	4	8	inverse pair
	3	3	9	9	9	9	9	9	9	9	absorbing
	4	4	7	1	4	7	1	4	7	1	inverse pair
	5	5	7	8	4	2	1	5	7	8	
	6	6	9	9	9	9	9	9	9	9	absorbing
	7	7	4	1	7	4	1	7	4	1	alternating
	8	8	1	8	1	8	1	8	1	8	
	9	9	9	9	9	9	9	9	9	9	absorbing

Figure 1: The canonical base-10 DRPT tile. Rows are residues n , columns are exponents k , and entries are $n^k \pmod{9}$, with residue 0 displayed as 9. The tile is the object: identity, inverse-pair rotation, cycle periodicity, and absorbing strata are directly visible and algebraically forced.

The displayed matrix is

$n \backslash k$	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1
2	2	4	8	7	5	1	2	4	8
3	3	9	9	9	9	9	9	9	9
4	4	7	1	4	7	1	4	7	1
5	5	7	8	4	2	1	5	7	8
6	6	9	9	9	9	9	9	9	9
7	7	4	1	7	4	1	7	4	1
8	8	1	8	1	8	1	8	1	8
9	9	9	9	9	9	9	9	9	9

This is not a sample or heuristic illustration. It is the complete finite object $D_9(n, k)$ for $1 \leq n, k \leq 9$.

4 Coordinate taxonomy

A cell of the tile is classified by the following coordinates.

4.1 Primary coordinates

The primary coordinates are

$$d, x, k, T_d(x, k).$$

Here d is the modulus, $x \in R_d$ is the row coordinate, $k \geq 1$ is the exponent coordinate, and $T_d(x, k) = x^k \bmod d$ is the cell value.

4.2 GCD-depth coordinate

Define the gcd-depth of a row element x by

$$\text{gd}(x) = \text{gcd}(x, d).$$

The unit stratum is $\text{gd}(x) = 1$. The non-unit strata are $\text{gd}(x) > 1$. This partitions the tile into structural row families.

Figure 2 shows the canonical tile colored by gcd-depth.

4.3 Prime-power valuation profile

Let

$$d = \prod_{i=1}^r p_i^{a_i}.$$

For a residue x , define its valuation profile

$$\nu_d(x) = (\min(a_i, v_{p_i}(x)))_{i=1}^r.$$

This profile controls how powers of x move through non-unit strata.

	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1
2	2	4	8	7	5	1	2	4	8
3	3	9	9	9	9	9	9	9	9
4	4	7	1	4	7	1	4	7	1
5	5	7	8	4	2	1	5	7	8
6	6	9	9	9	9	9	9	9	9
7	7	4	1	7	4	1	7	4	1
8	8	1	8	1	8	1	8	1	8
9	9	9	9	9	9	9	9	9	9

■ $\gcd(n, 9) = 1$: unit stratum
■ $\gcd(n, 9) = 3$: non-unit stratum
■ $\gcd(n, 9) = 9$: zero stratum

Figure 2: GCD-depth strata in the base-10 tile. Unit rows, non-unit rows, and the zero stratum are visually distinct because their arithmetic behavior is distinct.

4.4 Quotient-unit coordinate

If $g = \gcd(x, d)$, write

$$x = gu,$$

where u is coprime to d/g . Thus u defines a unit coordinate modulo

$$q = d/g.$$

This gives the quotient-unit coordinate

$$(x, d) \mapsto (g, q, u),$$

where

$$g = \gcd(x, d), \quad q = d/g, \quad u = x/g \pmod{q}.$$

This coordinate explains why two cells with the same displayed residue may nevertheless occupy different structural positions.

4.5 CRT coordinate

Using the Chinese Remainder Theorem,

$$\mathbb{Z}/d\mathbb{Z} \cong \prod_{i=1}^r \mathbb{Z}/p_i^{a_i}\mathbb{Z}.$$

A tile cell therefore has componentwise form

$$T_d(x, k) \leftrightarrow (x_i^k \bmod p_i^{a_i})_{i=1}^r.$$

Every row is a synchronized product of prime-power row behaviors.

5 The row law

Theorem 2 (Unit row periodicity). *Let $x \in R_d^\times$. Let*

$$p = \text{ord}_d(x)$$

be the multiplicative order of x modulo d . Then

$$T_d(x, k + p) = T_d(x, k)$$

for every $k \geq 1$. Moreover,

$$p \mid \lambda(d),$$

where $\lambda(d)$ is the Carmichael exponent of $R_d^\times$.

Proof. Since $p = \text{ord}_d(x)$, we have $x^p \equiv 1 \pmod{d}$. Therefore

$$x^{k+p} = x^k x^p \equiv x^k \pmod{d}.$$

The order of any element of a finite group divides the exponent of that group. Hence $p \mid \lambda(d)$. $\square$

Corollary 3 (Unit rows tile the exponent axis). *Every unit row is an infinite repetition of a finite word whose length divides $\lambda(d)$.*

6 The inverse pair law

Theorem 4 (Inverse row rotation). *Let $x \in R_d^\times$ have order p . Then the row of x^{-1} is a cyclic reversal of the row of x . More precisely,*

$$(x^{-1})^k \equiv x^{p-k} \pmod{d}.$$

Proof. Since $x^p \equiv 1 \pmod{d}$, we have $x^{-k} \equiv x^{p-k} \pmod{d}$. But $(x^{-1})^k = x^{-k}$. Therefore $(x^{-1})^k \equiv x^{p-k} \pmod{d}$. $\square$

Example 5 (Rows 2 and 5 in $d = 9$). In $R_9^\times$, $2^{-1} \equiv 5 \pmod{9}$. The row of 2 is

$$2, 4, 8, 7, 5, 1,$$

and the row of 5 is

$$5, 7, 8, 4, 2, 1.$$

These are inverse-cycle rotations.

Example 6 (Rows 4 and 7 in $d = 9$). In $R_9^\times$, $4^{-1} \equiv 7 \pmod{9}$. The row of 4 is

$$4, 7, 1, 4, 7, 1, \dots,$$

and the row of 7 is

$$7, 4, 1, 7, 4, 1, \dots$$

Again, inverse structure is visible as row rotation.

Figure 3 summarizes the row families of the canonical tile.

identity	rows: 1 1 → 1
full-order inverse pair	rows: 2, 5 2→4→8→7→5→1, 5→7→8→4→2→1
order-3 inverse pair	rows: 4, 7 4→7→1, 7→4→1
order-2 alternating	row: 8 8→1→8
absorbing zero-stratum	rows: 3, 6, 9 3→9, 6→9, 9→9

Figure 3: Row family taxonomy for the canonical tile. The visual families are the arithmetic families: identity, inverse pairs, order-2 alternation, and absorbing zero-stratum rows.

7 The non-unit law

Theorem 7 (Prime-power valuation law). *Let $d = p^a$. Let $x \in \mathbb{Z}/p^a\mathbb{Z}$, and suppose $v_p(x) = t$. If $t = 0$, then x is a unit. If $t \geq 1$, then*

$$v_p(x^k) = \min(a, kt).$$

In particular,

$$x^k \equiv 0 \pmod{p^a}$$

if and only if $kt \geq a$. The minimal hitting time to zero is

$$K_{p^a}(x) = \left\lceil \frac{a}{t} \right\rceil.$$

Proof. Write $x = p^t u$, with $p \nmid u$. Then $x^k = p^{kt} u^k$. Modulo p^a , the valuation is saturated at a . Therefore $v_p(x^k) = \min(a, kt)$. The residue is zero modulo p^a precisely when $p^a \mid x^k$, i.e. $kt \geq a$. $\square$

Theorem 8 (CRT non-unit classification). *Let*

$$d = \prod_{i=1}^r p_i^{a_i}.$$

A residue $x \in R_d$ reaches zero modulo d under powering if and only if

$$p_i \mid x$$

for every prime $p_i \mid d$. Equivalently,

$$\text{rad}(d) \mid x.$$

Proof. By CRT, $R_d \cong \prod_i R_{p_i^{a_i}}$. The residue x^k is zero modulo d if and only if it is zero in every prime-power component. By the prime-power valuation law, this occurs in component $p_i^{a_i}$ if and only if $p_i \mid x$. Thus x reaches zero globally if and only if every prime divisor of d divides x . $\square$

Corollary 9 (Non-unit permanence). *If $\gcd(x, d) > 1$, then $\gcd(x^k, d) > 1$ for all $k \geq 1$. Thus a non-unit row never returns to the unit group.*

Proof. If $p \mid x$ and $p \mid d$, then $p \mid x^k$ for all $k \geq 1$. Hence $\gcd(x^k, d) \geq p > 1$. $\square$

8 The column law

Rows flow. Columns act. For a fixed exponent k , the column map is

$$P_k : R_d \rightarrow R_d, \quad P_k(x) = x^k.$$

Theorem 10 (Column action on units). *Restricted to the unit group $R_d^\times$, the map $P_k(x) = x^k$ is a group endomorphism. Its image is*

$$(R_d^\times)^k = \{x^k : x \in R_d^\times\}.$$

When $R_d^\times$ is cyclic of order m , the image size is

$$\frac{m}{\gcd(k, m)}.$$

Proof. For units x, y , $P_k(xy) = (xy)^k = x^k y^k = P_k(x)P_k(y)$ because the group is abelian. If $R_d^\times$ is cyclic with generator g , then every unit is g^j , and $P_k(g^j) = g^{kj}$. The set of exponents $kj \pmod{m}$ has size $m/\gcd(k, m)$. $\square$

9 Row-column duality

Theorem 11 (Row-column duality). *The DRPT tile simultaneously encodes every row orbit $k \mapsto x^k$ and every exponent action $x \mapsto x^k$. Thus the tile is the finite incidence object of powering in R_d .*

Proof. By definition, $T_d(x, k) = x^k$. Fixing x gives a function of k . Fixing k gives a function of x . The same cell set supports both families. $\square$

10 The finite codebook theorem

The tile extends into infinity, but it does so finitely.

Theorem 12 (Eventual periodicity of the exponent axis). *Let $d = \prod_i p_i^{a_i}$ and let*

$$A(d) = \max_i a_i.$$

Then for every $x \in R_d$,

$$x^{k+\lambda(d)} \equiv x^k \pmod{d}$$

for all $k \geq A(d)$.

Proof. Work componentwise modulo $p_i^{a_i}$. If $p_i \nmid x$, then x is a unit modulo $p_i^{a_i}$, so $x^{\lambda(d)} \equiv 1 \pmod{p_i^{a_i}}$, because $\lambda(d)$ is divisible by the local Carmichael exponent. Hence $x^{k+\lambda(d)} \equiv x^k$. If $p_i \mid x$, then for $k \geq a_i$, the valuation law gives $x^k \equiv 0 \pmod{p_i^{a_i}}$, and likewise $x^{k+\lambda(d)} \equiv 0 \pmod{p_i^{a_i}}$. Since $k \geq A(d) \geq a_i$, both cases agree in every prime-power component. CRT completes the proof. $\square$

Corollary 13 (Finite codebook). *The infinite tile $T_d(x, k)$ is determined by the finite set of columns $1 \leq k \leq A(d) + \lambda(d)$. After the finite transient region, all rows repeat with period dividing $\lambda(d)$.*

This is the precise meaning of *finite object, infinite extension*. The tile is finite, but its exponent direction extends into infinity by a finite codebook.

11 CRT decomposition and cross-base atlas

Composite tiles are not opaque. They are synchronized products of prime-power component tiles. Figure 4 displays this for $d = 12$.

[Figure placeholder: CRT decomposition of $d = 12$ as $d = 3 \times d = 4$. Three side-by-side heatmaps showing tile mod 12, tile mod 3, and tile mod 4. Reproduce by computing T_{12} , T_3 , T_4 and rendering each as a heatmap.]

Figure 4: CRT decomposition of a composite tile. A composite modulus tile is a synchronized product of its prime-power component tiles.

The construction is uniform in d . In base b , the displayed DRPT is the case $d = b - 1$.

Theorem 14 (Base-covariant construction). *For every base $b \geq 2$, the associated DRPT tile is the power-residue tile of the modulus $d = b - 1$. All row laws, column laws, inverse-pair laws, non-unit laws, CRT decomposition laws, and finite-codebook laws hold uniformly in d .*

Proof. Each theorem above was stated for an arbitrary modulus d . Substituting $d = b - 1$ gives the base- b DRPT. No proof depends on the special value $b = 10$. $\square$

[Figure placeholder: Cross-modulus tile atlas. An 8-panel grid showing finite tiles for $d = 5, 6, 7, 8, 9, 10, 11, 12$ (i.e. bases 6 through 13). Each panel is a heatmap of $T_d(x, k)$. Reproduce by sweeping d over the range and rendering each tile.]

Figure 5: Cross-modulus tile atlas. Every base b gives a finite tile at $d = b - 1$. The display changes, while the structural laws remain uniform in d .

12 Display faithfulness

A visual display can mislead unless it is tied to exact predicates. Here the display is faithful.

Theorem 15 (Display faithfulness). *The displayed base- b DRPT tile $D_d(x, k)$ is a faithful presentation of the residue tile $T_d(x, k)$, up to the single convention identifying residue $0 \pmod{d}$ with the digit symbol d .*

Proof. The display map $\delta_d : R_d \rightarrow \{1, \dots, d\}$ is bijective if residue 0 is represented by d . Therefore every displayed entry uniquely determines the residue entry, and every residue entry has exactly one displayed representative. $\square$

The visual tile is not a heatmap imposed after the fact. It is a lossless display of the finite object, given the stated convention.

13 Line view vs. tile view

The ordinary number line shows a single ordering:

$$0, 1, 2, 3, \dots$$

The tile shows a two-coordinate action field:

$$(x, k) \mapsto x^k.$$

The line view collapses row orbit, exponent phase, gcd-depth, inverse family, CRT component, and column action. The tile restores them.

Proposition 16 (Loss of tile information under line projection). *The projection*

$$(x, k, T_d(x, k)) \mapsto T_d(x, k)$$

forgets row coordinate, exponent coordinate, and structural role. Therefore it is not injective.

Proof. In the base-10 tile,

$$T_9(2, 6) = 1 \quad \text{and} \quad T_9(8, 2) = 1.$$

The projected value is 1 in both cases, but the cells have different row and exponent coordinates. Therefore the projection is not injective. $\square$

14 Base-family audit protocol

Because theorems are proven algebraically, a base sweep is not needed as proof. It is included for reproducibility, atlas construction, and hostile-reader confidence.

For bases $b = 2, \dots, 901$, define $d = b - 1$. For each d , the audit checks tile existence and the following structural predicates:

- unit Carmichael identity;
- inverse row rotation;
- non-unit permanence;
- eventual periodicity after $A(d)$;
- factorization and Carmichael metadata for the base family.

The generated certificate reports:

$$\text{case count} = 900, \quad \text{pass count} = 900, \quad \text{fail count} = 0.$$

The certificate hash is

$$03f81de0d6e3bba963db52e5328007ea2a45d733d92cb83426b0ecb8afd4337f.$$

This audit is not the proof of the universal statements. The proof is algebraic. The audit is a reproducible coverage surface for 900 consecutive base instances.

15 Future directions

Several natural questions arise from the static framing developed here. First, the codebook bound $A(d) + \lambda(d)$ in Theorem 12 is uniform but not necessarily sharp; explicit witnesses showing that the transient region cannot be shortened in general would settle the question of optimality. Second, the column endomorphisms $P_k(x) = x^k$ generate a multiplicative semigroup acting on R_d ; its spectral structure relative to the additive Fourier transform on $\mathbb{Z}/d\mathbb{Z}$ is a natural target for further analysis, paralleling the additive setting of [3, 4]. Third, the CRT decomposition of Section 8 suggests treating the family $\{T_d\}_{d \geq 1}$ as a projective system under base-change maps; a systematic study of compatible morphisms across moduli is left for future work.

16 Conclusion

The power-residue tile is a faithful finite arithmetic object.

For every modulus d , the tile

$$T_d(x, k) = x^k \pmod{d}$$

is finite, exhaustible, and fully determined by residue arithmetic. It displays unit cycles, inverse-pair rotations, non-unit valuation behavior, CRT decomposition, column action, and finite-codebook extension.

For base b , the digital-root display is the case $d = b - 1$. The base-10 9×9 tile is not an illustration. It is the complete finite object for $d = 9$. Its visible rows are lawful: row 1 is identity; rows 2 and 5 are inverse full-order cycles; rows 4 and 7 are inverse order-3 cycles; row 8 is order 2; and rows 3, 6, and 9 occupy the absorbing zero-stratum under the digital-root convention.

The tile is finite, but its exponent direction extends into infinity by a finite codebook. The structural laws stated above are uniform in d and elementary in their proofs. The visual presentation, given the digital-root display convention, is a lossless rendering of the underlying residue arithmetic.

A Base-10 tile data

$n \backslash k$	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1
2	2	4	8	7	5	1	2	4	8
3	3	9	9	9	9	9	9	9	9
4	4	7	1	4	7	1	4	7	1
5	5	7	8	4	2	1	5	7	8
6	6	9	9	9	9	9	9	9	9
7	7	4	1	7	4	1	7	4	1
8	8	1	8	1	8	1	8	1	8
9	9	9	9	9	9	9	9	9	9

B Base-10 row taxonomy

For $d = 9$, $\lambda(9) = 6$, and

$$R_9^\times = \{1, 2, 4, 5, 7, 8\}.$$

Row	$\gcd(n, 9)$	Behavior	Family
1	1	order 1	identity
2	1	order 6	full-order unit
3	3	reaches zero class	non-unit
4	1	order 3	order-3 unit
5	1	order 6	inverse of 2
6	3	reaches zero class	non-unit
7	1	order 3	inverse of 4
8	1	order 2	order-2 unit
9	9	zero class	zero-stratum

C Minimal reproducibility script

```
def drpt_tile(d, K=None):
    if K is None:
        K = d
    def show(r):
        return d if r % d == 0 else r % d
    return [[show(pow(n, k, d)) for k in range(1, K+1)]
            for n in range(1, d+1)]

tile = drpt_tile(9, 9)
for row in tile:
    print(row)
```

Expected output:

```
[1, 1, 1, 1, 1, 1, 1, 1, 1]
[2, 4, 8, 7, 5, 1, 2, 4, 8]
[3, 9, 9, 9, 9, 9, 9, 9, 9]
[4, 7, 1, 4, 7, 1, 4, 7, 1]
[5, 7, 8, 4, 2, 1, 5, 7, 8]
[6, 9, 9, 9, 9, 9, 9, 9, 9]
[7, 4, 1, 7, 4, 1, 7, 4, 1]
[8, 1, 8, 1, 8, 1, 8, 1, 8]
[9, 9, 9, 9, 9, 9, 9, 9, 9]
```

References

- [1] T. M. Apostol, *Introduction to Analytic Number Theory*, Undergraduate Texts in Mathematics, Springer, 1976.
- [2] W.-S. Chou and I. E. Shparlinski, *On the cycle structure of repeated exponentiation modulo a prime*, J. Number Theory **107** (2004), 345–356.
- [3] J. Grieshop, *Explicit constants for Fejér-type smoothing on finite cyclic groups*, arXiv:2512.03085 [math.CA], 2025.
- [4] J. Grieshop, *On the failure of step-response tests to certify admissibility of spectral averaging operators*, arXiv:2602.20185 [math.CA], 2026.
- [5] J. Grieshop, *Marithmetics (archival release)*, Zenodo, 2026. doi:10.5281/zenodo.19424837.
- [6] J. Grieshop, *Marithmetics: source repository, scripts, and demonstrations*, GitHub repository, 2026. <https://github.com/public-arch/Marithmetics>.
- [7] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., revised by D. R. Heath-Brown and J. H. Silverman, Oxford University Press, 2008.
- [8] I. M. Izmirli, *On some properties of digital roots*, Advances in Pure Mathematics **4** (2014), 295–301.
- [9] T. D. Rogers, *The graph of the square mapping on the prime fields*, Discrete Math. **148** (1996), 317–324.

- [10] L. DeVille, *The shape of $x^2 \bmod n$* , arXiv:2207.10512 [math.CO], 2022.
- [11] T. Vasiga and J. Shallit, *On the iteration of certain quadratic maps over $\text{GF}(p)$* , Discrete Math. **277** (2004), 219–240.